Elvis F. Elli

From: catherine.bohnert@usda.gov
Sent: Thursday, August 20, 2026 1:10 PM
To: Elvis F. Elli
Subject: Review Package for Proposal 2025-09471 Submitted to USDA\NIFA

Area: Agriculture and Food Research Initiative
Program: Foundational Knowledge of Agricultural Production Systems
Proposal Number: 2025-09471
Project Director: Elvis Elli
Proposal Title: Agronomic and physiological strategies to maximize soybean systems performance to drought and high-temperature risks

Dr. Elvis Elli
Division of Agriculture of the University of Arkansas System
1354 W. Altheimer Dr.
Fayetteville, AR 72704-6898

Dear Dr. Elli,

Thank you for submitting your application to the National Institute of Food and Agriculture (NIFA) Agriculture and Food Research Initiative (AFRI) - Foundational Knowledge of Agricultural Production Systems (A1102) program. We regret to inform you that your application was not recommended for funding.

The review panel members carefully evaluated your application. These panel members are fellow scientists selected for their competence in research, education, and extension, and their knowledge of your field. Your application was evaluated on scientific merit and the quality of your application using the evaluation criteria associated with the FY 2026 AFRI Foundation and Applied Science Program Request for Applications.

Please consult the AFRI Foundational and Applied Science Program NOFO once it is published to align your project with the appropriate priority area. We encourage any unsuccessful applicant to carefully consider the comments of the peer review panel if they wish to reapply.

You will find the panel summary, the individual reviews, and the relative ranking sheet below. We hope you find this information useful. If you have questions after you have had the opportunity to read and evaluate the individual reviews, please do not hesitate to contact Dr. Mat Ngouajio or Dr. John Erickson after October 1, 2026, to set up an appointment to discuss.

Thank you for your interest in the AFRI Foundational Knowledge of Agricultural Production Systems program. We wish you all the best in future application submissions.

Sincerely,

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United States Department of Agriculture
National Institute of Food and Agriculture
Agriculture and Food Research Initiative
Foundational Knowledge of Agricultural Production Systems

Proposal 2025-09471 submitted by Elli

Agronomic and physiological strategies to maximize soybean systems performance to drought and high-temperature risks

The review panel grouped proposals into one of the relative categories below. The percentage indicates the final distribution of proposals in each category.

Outstanding %	9
High Priority %	29
Medium Priority %	31
Low Priority %	28
Do Not Fund %	3

This proposal was placed in : Medium Priority

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Foundational Knowledge of Agricultural Production Systems - PANEL SUMMARY

The panel decision regarding your proposal is based on the input provided by the reviews and the collected expertise and judgment of the individual panel members. This panel summary reflects the consensus opinion of the panel regarding your proposal.

Proposal Number: 2025-09471   Project Director: Elli

Proposal Title: Agronomic and physiological strategies to maximize soybean systems performance to drought and high-temperature risks

Positive Aspects of the Proposal
The proposed work aims to integrate physiological responses of soybean to drought into predictive crop-system models to improve predictive power and then use that improved model to develop and implement strategies to promote soybean production under variable weather conditions. Some valuable data will be generated from the work by measuring on-farm soybean seasonal crop evapotranspiration and crop growth dynamics. The proposed work is novel and if successful would improve model performance and predictive capacity.

Negative Aspects of the Proposal
Although the approach and methodology in Objective 1 were quite extensive, it is not as clear in Objective 2, particularly on how a new crop model system developed in Objective 1 could be applied to Objective 2. There are insufficient descriptions on the experimental design, such as how many leaves per plant, or the location on the plant, that Li-6800 data will be collected. It would be good to record VPD at multiple critical growth stages. There is also no mention of weed-crop competition. Of the 10 sites in Objective #2, eight sites have weather stations, but no information for the remaining two sites on how/where to collect current and/or historical weather data. It is also missing detail on statistical analyses across all objectives.

Synthesis Comments
This proposal intends to develop a new crop model system at the leaf level by improving management and physiological strategies to increase soybean yield. The proposed work is novel and if successful it would improve model performance and predictive capacity. However, some consideration in detail on the experimental design would strengthen the proposal.

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The following reviews were submitted for your proposal, the names of the reviewers have been removed to maintain confidentiality.
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Project Relevance

Strengths:

Objective 1 data will be used for DCaPST model parameterization, with the potential to identify genotypes with an enhanced ability to conserve water under stressed environments. This data will help support, but does not limit, the goal of objective 2 to use an improved and validated version of a cropping system model to run a what-if scenario analysis to identify sustainable solutions for adapting soybean systems to local weather fluctuations.

Weaknesses:

Scientific Merit- Innovation & Approach

Strengths:

Objective 1, Task 1.2 will provide valuable data by measuring on-farm soybean seasonal crop evapotranspiration and crop growth dynamics.

Weaknesses:

Weed-crop competition drastically alters water availability, soybean growth, and soybean yield. Year-to-year weather fluctuations influence weed competition, for example, weed-crop competition may differ from a year with a wet spring that supports a large flush of weed emergence as compared to a season-long drought, with species-level difference based on competitive ability, drought tolerance, C3/C4, herbicide-resistance status, and more. There is not a single mention of weed-crop competition, which should at least be acknowledged as a pitfall, or briefly mentioned as a major factor in the introduction.

Objective 1, Task 1.1: No mention of how many leaves per plant, or the location on the plant, that Li-6800 data will be collected, with this data collection being a tedious process.

Objective 1, Task 1.1: Plants will be well-watered until R1 until the VPD measurement date. Would be beneficial to record VPD and multiple critical growth stages.

Objective 2, Task 2.1: Of the 10 sites, 8 have weather stations. No information on how/where current and/or historical weather data will be pulled for the remaining 2 sites.

Objective 2, Task 2.2: Will run APSIM simulations over at least 40 historical weather years (1985-2024) to account for year-to-year weather variability. Wondering if any thought was put into how the adoption of RR soybeans in 1996 onward may influence the model.

Qualifications of investigators and facilities: The investigator team appears to have the requisite experience and facilities to complete this project.

Data Management Plan: The data management plan appears complete.

Response to Previous Reviews: The proposal has been revised to address most previous reviewer concerns, including a focus on novelty, reframing the experimental data collection, provided details on model paramaterization. Details related to stakeholder engagement with outcomes are still sparse.

Relevance to RFA: The proposed work addresses A1102 priority areas investigating (1b) how cropping system performance is impacted by management and (1c) extending models to derive general principles of crop performance.

Innovation: The proposed work aims to integrate physiological responses of soybean to drought into predictive crop-system models to improve predictive power and then use that improved model to develop and implement strategies to promote soybean production under variable weather conditions. The proposed work is novel and if successful would improve model performance and predictive capacity. The investigators have preliminary data supporting variation among soybean varietals in their response to altered VPD, which supports their hypothesis that variation in physiological responses to climate variability impact WUE across soybean genotypes. Additionally, preliminary simulation exercises support the potential value of the proposed work.

Approach: The proposed work will consist of growth chamber and field experiments to collect physiological data, use that data to paramaterize the DCaPST model and ultimately incorporated into the AGSIM model, and then test the model using historical data. The validated model will be used to develop scenarios and provide management options for producers, to be disseminated using an RShiny app. The methodological approach is generally well-described. A weakness is a lack of detail related to statistial analyses to be conducted across all objectives. Another minor weakness is the binning of historical data into categories of normal, wet/warm, etc years, rather than using the data as continuous variables (perhaps aggregating them using an ordination approach). Project results will be disseminated through publications/presentations, incorporated into course material, and presented to the public via the RShiny app. More details on stakeholder engagement with results/outcomes would have strengthened the proposal.

Qualifications: The investigators are well-qualified to conduct the proposed research (including physiology, modeling, and fluxes) and have the necessary facilities and equipment.

Data Management Plan: The data management plan is adequate.

Project Management Plan: The project management plan is thorough and achievable, including an advisory board.

Mentoring Plan: Both the investigators are experienced mentors and have undergone mentoring training. The mentoring plan is comprehensive and contains specific details tailored to undergrad/grad/postdoc career stages related to both research skill and professional development training.

Overall: The proposed project addresses an important knowledge gap in soybean agriculture and would serve to improve management applications to promote soybean production. The approach is mostly clear, and there is a strong foundation of project management, data management, and mentoring plans in place to promote project success.

Strength

With increasing impact on crop production from environmental stresses, such as extreme drought and high temperature, there is an urgent need to develop an improved crop model for soybean that can operate at the leaf level to simulate soybean growth, seed yield and water use. In addition, through extensive evaluation of the year-to-year weather fluctuations, soybean seed yield and water use efficiency, they intend to identify the most effective management and physiological strategies that would increase soybean yield. They propose to advance current soybean model capabilities by implementing new algorithms within a cropping system model (Agricultural Production Systems Simulator – APSIM) that simulate leaf-level photosynthesis, respiration, and stomatal conductance processes using the Diurnal Canopy Photosynthesis–Stomatal Conductance (DCaPST) model.

Weakness

This project is mainly focused on a regional study on the effect of drought and high temperature on soybeans in Arkansas. It is not clear how it will be applied for other soybean growing regions in the country.

Can this model be adapted in other soybean growing states? This research is focused mainly on Arkansas soybean.

It is not clear who owns the APSIM framework and DCaPST model? it is not clear how can they be accessed by the U.S. growers?

Although the approach and methods in Objective 1 to conduct experiments in controlled environment were quite extensive, it is not as clear in Objective 2, particularly on how a new crop model system developed in Objective 1 could be applied for Objective 2.

Conclusion

This proposal intends to develop a new crop model system at the leaf level by improving management and physiological strategies to increase soybean yield in Arkansas.